

Software Requirements Specification (SRS)

Gym Management System

Prepared for: **Nejat**

Prepared by: **Nafiyad**

Date: **December 28, 2024**

Table of Contents

1. Introduction
2. System Requirements
3. System Design Overview
4. Modules and Features
5. Appendices

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for the Gym Management System. The system aims to streamline gym operations by providing functionalities such as **membership management, billing, class scheduling, trainer management, and more.**

1.2 Scope

The Gym Management System will serve gym administrators, trainers, and members by integrating modules like Membership Management, Billing and Payment, Class Scheduling, Trainer and Staff Management, Reporting and Analytics, Mobile App Integration, Inventory Management, and Facility and Equipment Management. The system will also manage staff commissions and provide time-based subscription options.

1.3 Definitions, Acronyms, and Abbreviations

- **Membership:** A registered gym user with a subscription.
- **Trainer:** Gym staff who provide training services to members.
- **Subscription:** A predefined time-based gym membership plan.
- **SRS:** Software Requirements Specification.

2. System Requirements

2.1 Functional Requirements

The system should meet the following functional requirements for each module:

1. Membership Management

- Member registration with personal details and membership plans.
- Subscription management, including renewals, upgrades, and cancellations.
- Automatic alerts for upcoming renewals and expired memberships.
- Reports on active, inactive, and cancelled memberships.

2. Billing and Payment Management

- Support for multiple payment methods (cash and online).
- Invoice generation and email notifications.
- Track subscription payments and pending dues.
- Refund processing for cancellations.

3. Class and Schedule Management

- Class creation and scheduling with capacity limits.
- Booking and cancellation by members through the app.
- Trainer availability tracking for scheduling.
- Conflict resolution for overlapping schedules.

4. Trainer and Staff Management

- Trainer profiles with qualifications and specializations.
- Scheduling for personal training sessions.
- Performance tracking and feedback collection.
- Calculation and tracking of income percentages from training sessions.

5. Reporting and Analytics

- Membership trend analysis (new vs. renewals).
- Financial performance reports (income vs. expenses).
- Inventory and equipment usage statistics.

6. Mobile App Integration

- Member self-service for class bookings and cancellations.
- Subscription and payment history tracking.
- Push notifications for class reminders and renewals.

7. Inventory Management

- Tracking gym merchandise (e.g., supplements, apparel).
- Stock alerts for low inventory.
- Sales tracking and reporting.
- Integration with billing for merchandise purchases.

Additional Features

- Staff commission tracking based on training sessions.
- Time-based subscriptions with automatic expiration.
- Optional 10-day extension with valid reason submission and approval.

3. System Design Overview

The system design will follow a modular architecture, integrating all functionalities as distinct modules.

3.1 Architectural Design

Details of the architecture will be provided during the design phase.

3.2 Data Flow and Use Case Diagrams

Visual representations will be included in the design phase.